Universal Baseline Law: A Unifying Framework for Physical Noise Floors Across Classical and Quantum Domains

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Abstract

A remarkable structural regularity pervades the noise floors of disparate physical systems: from resistors and capacitors to mechanical oscillators, radio-frequency channels, magnetic sensors, accelerometers, and quantum vacuum boundaries, the irreducible baseline fluctuation in every observable takes the form of a square root of a ratio of thermal (or zero-point) energy, accessible degrees of freedom, and a constraining parameter. We term this the **Universal Baseline Law** and express it as:

$$\sigma \propto \sqrt{(E_{\text{scale}} \cdot N_{\text{freedom}} / \kappa_{\text{constraint}})}$$

where E_{scale} is the relevant energy scale ($k_B T$ classically, $\hbar\omega/2$ in the quantum regime), N_{freedom} is the number of active degrees of freedom, and $\kappa_{\text{constraint}}$ is the stiffness, capacitance, permeability, or analogous restoring parameter of the system. We demonstrate that Johnson–Nyquist noise, kT/C capacitor noise, RF thermal noise, Brownian motion, thermomechanical accelerometer noise, magnetic Johnson noise, and Casimir force fluctuations are all instances of this single law under appropriate variable substitution. The derivation proceeds from the equipartition theorem and the fluctuation–dissipation theorem, and the square-root structure is shown to follow necessarily from the isotropy of thermal fluctuations in phase space. Practical implications for sensor design, quantum-limited measurement, and noise engineering are discussed.

1. Introduction

Across every branch of experimental physics and precision engineering, an inescapable observation recurs: no matter how carefully an instrument is constructed, how meticulously electromagnetic interference is shielded, or how cleverly the readout circuit is optimized, a residual fluctuation persists at the system's output. This irreducible agitation is not the product of poor craftsmanship. It is the signature of thermodynamics itself — a fundamental noise floor imposed by the finiteness of temperature and the quantization of action. Remarkably, despite manifesting in systems as structurally dissimilar as a carbon-film resistor, a MEMS accelerometer, an RF receiver front end, and the vacuum gap between two conducting plates, these floors share a single, unmistakable mathematical signature: the *square root of a product of energy and freedom over constraint*. The present paper formalizes this observation as the Universal Baseline Law (UBL) and demonstrates, through rigorous derivation and explicit domain mapping, that it is not a coincidence but a mathematical necessity.

The history of noise physics is a century-long accumulation of insights that, in retrospect, all point toward the same structure. Robert Brown's 1828 microscopic observations of the incessant, irregular motion of pollen grains suspended in water [17] provided the first empirical window onto thermal fluctuations, though a quantitative theory would wait three-quarters of a century. Albert Einstein's foundational 1905 paper [3] derived the diffusion constant for a Brownian particle from purely statistical-mechanical arguments, establishing that the mean-squared displacement grows linearly in time with coefficient $D = k_B T / (m\gamma)$ — a quantity entirely determined by temperature, mass, and damping. Just over two decades later, John B. Johnson (1928) [1] measured the spontaneous voltage fluctuations across a resistor in thermal equilibrium, and Harry Nyquist (1928) [2] simultaneously provided the theoretical explanation via a one-dimensional electromagnetic transmission-line argument and the equipartition theorem, establishing the power spectral density $S_V(f) = 4k_B TR$. The equipartition theorem itself — the proposition that each quadratic degree of freedom in the Hamiltonian carries mean thermal energy $k_B T/2$ at equilibrium — was formalized by Boltzmann and Maxwell in the late nineteenth century [7], and stands as the most direct ancestor of the UBL.

The theoretical synthesis achieved by Callen and Welton in 1951 [4] — the fluctuation–dissipation theorem (FDT) — demonstrated that any dissipative mechanism in a system at thermal equilibrium is necessarily accompanied by spontaneous fluctuations of precisely calibrated magnitude, with the two phenomena governed by the same susceptibility function. This theorem unified Johnson noise, Brownian motion, and their quantum generalizations under a single theoretical roof. Independently, Hendrik Casimir's 1948 prediction [5] of an attractive force between two perfectly conducting parallel plates, arising from the modification of zero-point electromagnetic vacuum fluctuations, extended the domain of physical fluctuations into the purely quantum-mechanical regime. Kubo's subsequent development [6] of linear response theory formalized the FDT in a general many-body setting, and the work of Braginsky and Khalili [11] extended these ideas to quantum measurement theory and the Standard Quantum Limit.

Yet despite this century of progress, no single closed-form expression with explicit, quantitative mappings across all major physical noise domains has been stated and proven as a *universal* law. Individual noise formulas are taught in isolation — Johnson noise in electronics courses, Brownian motion in statistical mechanics, thermomechanical noise in MEMS textbooks, Casimir fluctuations in quantum field theory — and the structural identity connecting them is rarely if ever made explicit. The contribution of the present paper is precisely this: to state the Universal Baseline Law in closed form, derive it from first principles with full generality, map seven major physical domains explicitly onto its three variables, and demonstrate through a geometric isotropy argument that the square-root structure of the law is not arbitrary but is forced by the quadratic nature of physical Hamiltonians near equilibrium and the resulting isotropy of the thermal phase-space distribution.

The practical motivation for this synthesis is substantial and growing. As sensors — gyroscopes, accelerometers, magnetometers, gravimeters, RF receivers, and optical interferometers — are pushed ever closer to their fundamental limits, the distinction between engineering imperfection and thermodynamic inevitability becomes operationally critical. LIGO's gravitational-wave detectors operate within a factor of a few of the Standard Quantum Limit [15], MEMS inertial sensors are thermomechanically limited at room temperature [8], and next-generation NEMS devices at nanometer separations are entering the regime where Casimir force fluctuations become a dominant noise source [14]. The UBL provides a unified framework for computing, comparing, and engineering these limits across all such systems.

The remainder of the paper is organized as follows. Section 2 reviews the three theoretical pillars — the equipartition theorem, the fluctuation–dissipation theorem, and phase-space geometry — that together support the UBL. Section 3 surveys the noise laws of seven distinct physical domains and identifies the UBL structure in each. Section 4 provides a general derivation of the UBL from first principles. Section 5 analyzes the geometric origin of the square-root structure. Section 6 presents the isotropy argument. Section 7 provides an explicit domain-mapping table. Section 8 discusses practical implications for sensor design and quantum-limited measurement. Section 9 concludes with a summary and outlook.

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2. Theoretical Foundations

The Universal Baseline Law rests on three interlocking theoretical pillars: the equipartition theorem of classical statistical mechanics, the fluctuation–dissipation theorem of linear response theory, and the geometric structure of phase-space distributions for quadratic Hamiltonians. Each pillar independently constrains the form of noise amplitudes, and together they uniquely determine the UBL.

2.1 The Equipartition Theorem

For a physical system in thermal equilibrium with a heat reservoir at absolute temperature T, the canonical ensemble of statistical mechanics assigns to each quadratic term in the Hamiltonian a mean energy of exactly $k_B T/2$, where $k_B = 1.380 \times 10^{-23}$ J/K is Boltzmann's constant [7, 16]. This is the equipartition theorem, and it applies identically to kinetic energy terms ($\frac{1}{2}m\dot{r}^2$) and potential energy terms ($\frac{1}{2}\kappa q^2$) alike, provided the system is in thermal equilibrium and the relevant degrees of freedom are active (i.e., the thermal energy $k_B T$ greatly exceeds the mode's quantum energy $\hbar\omega$).

Consider a generalized coordinate q — which may represent a displacement, a voltage, a charge, or any other physical observable — with a quadratic restoring Hamiltonian:

(1) $H = \frac{1}{2}\,\kappa\, q^2$

The equipartition theorem then gives:

(2) $\langle \frac{1}{2}\,\kappa\, q^2 \rangle = \frac{1}{2}\, k_B T \quad \Rightarrow \quad \langle q^2 \rangle = k_B T / \kappa$

Taking the square root, the root-mean-square (RMS) fluctuation in the observable q is:

(3) $\sigma_q = \sqrt{(\, k_B T / \kappa\,)}$

Equation (3) is the seed of the Universal Baseline Law. Every specific noise formula derived in subsequent sections will reduce to this form, with $k_B T$ replaced by the appropriate energy scale and κ replaced by the domain-specific constraint parameter. The power of the theorem lies in its universality: it makes no reference to the microscopic mechanism of fluctuation, only to the quadratic structure of the Hamiltonian and the fact of thermal equilibrium.

2.2 The Fluctuation–Dissipation Theorem

The fluctuation–dissipation theorem (FDT), proven in its general form by Callen and Welton in 1951 [4] and subsequently formalized by Kubo in the framework of linear response theory [6], establishes a precise and profound relationship between the spontaneous equilibrium fluctuations of a physical observable and the dissipative response of the same system when externally driven. In essence: any system that dissipates energy when driven by an external force must, by the requirement of detailed balance at thermal equilibrium, also exhibit spontaneous fluctuations of exactly the amplitude needed to maintain the equilibrium distribution.

Let x be a generalized observable and $\chi(\omega) = \chi'(\omega) + i\chi''(\omega)$ the complex generalized susceptibility (where $\chi''(\omega) = \mathrm{Im}[\chi(\omega)]$ is the dissipative part). The one-sided power spectral density of spontaneous fluctuations in x is given by the classical FDT as:

(4) $S_x(\omega) = (2k_BT / \omega)\ \mathrm{Im}[\chi(\omega)]$ [classical limit, $\hbar\omega \ll k_BT$]

In the full quantum case, the thermal factor is replaced by the quantum-corrected expression:

(5) $S_x(\omega) = \hbar \coth(\hbar\omega / 2k_BT) \cdot \mathrm{Im}[\chi(\omega)]$

which reduces to (4) in the limit $k_BT \gg \hbar\omega$ and to $\hbar\ \mathrm{Im}[\chi(\omega)]$ at $T = 0$ (the zero-point fluctuations). The FDT guarantees that any dissipation mechanism — electrical resistance, mechanical damping, magnetic eddy currents — is inseparably linked to an accompanying fluctuation, with the amplitude set by the same parameter. This is why the noise floor of any dissipative system is irreducible: one cannot eliminate the dissipation without simultaneously removing the signal-coupling mechanism. The theorem also confirms that the relevant energy scale is k_BT classically and $\hbar\omega/2$ at the quantum zero-point limit, precisely as asserted in the UBL.

2.3 Phase-Space Geometry and the Square Root

A third, more geometric perspective illuminates why the noise *amplitude* — and not the noise power — is the physically fundamental quantity and why it necessarily scales as a square root. For a system with a quadratic Hamiltonian and N degrees of freedom, the thermal equilibrium distribution in phase space is a multivariate Gaussian:

(6) $\rho(q, p) \propto \exp[\ -H(q, p) / k_BT\ ] = \exp[\ -\kappa q^2/(2k_BT) - p^2/(2mk_BT)\ ]$

The phase-space volume occupied at the 1σ level scales as $(2\pi)^N \prod_i \sigma_{qi}\, \sigma_{pi} = (2\pi\hbar)^N \times (k_B T/\hbar\omega)^N$, which grows as $(k_B T)^N$. The variance in any single coordinate, however, scales as $k_B T$ — a linear function of temperature. The *amplitude* (standard deviation) then scales as $\sqrt{(k_B T)}$. This square-root relationship is not a convenience or an approximation: it is the exact mathematical consequence of Gaussian statistics arising from quadratic Hamiltonians combined with the Boltzmann distribution. The Heisenberg uncertainty principle further requires that $\sigma_q\, \sigma_p \geq \hbar/2$, which the thermal distribution saturates only in the quantum ground state. These three considerations together establish both the functional form and the physical origin of the UBL.

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3. Cross-Domain Survey of Noise Laws

This section surveys seven major physical noise domains, deriving or citing the established noise formula in each case and identifying within each formula the three UBL variables: the energy scale E_{scale}, the freedom parameter N_{freedom}, and the constraint parameter $\kappa_{\text{constraint}}$. The identification is made explicit in the domain-mapping table of Section 7.

3.1 Johnson–Nyquist Noise (Resistors)

The most celebrated instance of thermal noise in electrical engineering is the voltage fluctuation spontaneously generated across a resistor R in thermal equilibrium at temperature T. Johnson (1928) [1] measured these fluctuations experimentally with extraordinary precision, and Nyquist (1928) [2] immediately provided the theoretical derivation by modeling the resistor as a one-dimensional electromagnetic transmission line and applying the equipartition theorem to each standing-wave mode. The one-sided power spectral density of the open-circuit voltage noise is:

(7) $S_V(f) = 4 k_B T R \quad [\text{V}^2/\text{Hz}]$

Over a measurement bandwidth Δf, the RMS noise voltage is obtained by integrating the spectral density:

(8) $V_{\mathrm{rms}} = \sqrt{(4k_B TR\,\Delta f)}$

The UBL variable identification is as follows. The energy scale is $E_{\mathrm{scale}} = k_B T$. The freedom parameter is $N_{\mathrm{freedom}} = 4R\Delta f$, which captures the product of the dissipative coupling strength (resistance) and the number of accessible frequency modes in the measurement band. The constraint parameter is $\kappa_{\mathrm{constraint}} = 1$ (dimensionally, the voltage is the natural observable and no additional restoring stiffness appears for a pure resistor, in contrast to the capacitor case). Equation (8) then reproduces equation (3) with N_{freedom} explicitly entering as a multiplier: $\sigma_V = \sqrt{(k_B T \cdot 4R\Delta f / 1)}$. The quantum correction replaces $k_B T$ with $(\hbar\omega/2)\coth(\hbar\omega/2k_B T)$ and is significant only at microwave or higher frequencies at room temperature.

3.2 Capacitor (kT/C) Noise

A result of remarkable elegance — and of great practical importance in sampled-data and analog circuit design — is the total integrated noise voltage across a capacitor C connected to any resistive network in thermal equilibrium. Regardless of the topology or value of the surrounding resistances, the total mean-square noise voltage is:

(9) $V_{\mathrm{rms}} = \sqrt{(k_B T / C)}$

This result follows directly and immediately from equipartition applied to the energy stored in the capacitor: $\langle \tfrac{1}{2}CV^2 \rangle = \tfrac{1}{2}k_B T$, giving $\langle V^2 \rangle = k_B T/C$. Resistance determines the *bandwidth* over which the noise is spread (via the RC time constant) but does not affect the total integrated noise power.

This is the cleanest and most transparent instance of the UBL (equation 3): $E_{\mathrm{scale}} = k_B T$, $N_{\mathrm{freedom}} = 1$ (a single electrical degree of freedom — the voltage across the capacitor), and $\kappa_{\mathrm{constraint}} = C^{-1}$... but conventionally written with C in the denominator directly, so $\kappa_{\mathrm{constraint}} \equiv 1/C$. The electrical stiffness (inverse capacitance, $1/C$) plays the role of the mechanical spring constant κ in an exact electrical-mechanical analogy. The kT/C noise floor is the binding fundamental limit on dynamic range in switched-capacitor circuits, analog-to-digital converters, charge-sensing amplifiers, and biosensors that rely on capacitive sampling [9].

3.3 RF Thermal Noise (Available Noise Power)

In a matched radio-frequency system at temperature T, the available thermal noise power per unit bandwidth deliverable from any resistive two-terminal source into a conjugate-matched load is:

(10) $P_N / \Delta f = k_B T \quad [\mathrm{W/Hz}]$

This result, which follows from the same Nyquist transmission-line argument as equation (7) applied to the power rather than the open-circuit voltage, is noteworthy for its independence from impedance: the available power $P_N = k_B T \Delta f$ does not depend on R. The corresponding RMS noise voltage across the source into a matched load of impedance R is:

(11) $V_{rms} = \sqrt{(k_B T R \Delta f)}$

which differs from equation (8) by a factor of 2 in the argument (the matched-load condition halves the available voltage). In the UBL framework: $E_{scale} = k_B T$, $N_{freedom} \propto \Delta f$ (the number of propagating electromagnetic modes within the bandwidth, each carrying $k_B T$ of energy in the classical regime), and $\kappa_{constraint} \sim 1/R$ (characteristic impedance, acting as a constraint on the coupling of electromagnetic energy to the circuit). The noise figure of RF amplifiers and receivers is defined relative to this fundamental floor [2, 9].

3.4 Brownian Motion

Brownian motion [17, 3] is the oldest experimental window onto thermal fluctuations and the most historically prominent instance of the UBL. For a particle of mass m confined in a harmonic potential with spring constant κ and in thermal equilibrium, the equipartition theorem directly gives the RMS displacement:

(12) $x_{rms} = \sqrt{(k_B T / \kappa)}$

For a free Brownian particle ($\kappa \to 0$) observed over a time interval τ much longer than the momentum relaxation time $\tau_c = 1/\gamma$ (where γ is the damping rate), the Einstein–Smoluchowski relation [3] gives:

$$(13)\ x_{rms} = \sqrt{(2D\tau)} = \sqrt{(2k_B T\tau / (m\gamma))}$$

where $D = k_B T / (m\gamma)$ is the translational diffusion constant. In equation (12), the UBL identification is immediate: $E_{scale} = k_B T$, $N_{freedom} = 1$ (single translational mode), $\kappa_{constraint} = \kappa$ (spring constant). In equation (13), the effective "freedom" is proportional to τ/τ_c (the number of independent random kicks received), and the constraint is the mechanical impedance $m\gamma$. In both regimes, the UBL structure is manifest. Einstein's derivation [3] of the diffusion constant from the Stokes drag force — $m\gamma = 6\pi\eta r$ for a sphere of radius r in a fluid of viscosity η — represents the first explicit use of what would later be recognized as the fluctuation–dissipation theorem.

3.5 Thermomechanical Accelerometer Noise

MEMS accelerometers and resonators are limited at room temperature by thermomechanical (Brownian) noise in their proof masses and suspension springs [8, 10]. The displacement noise spectral density of a damped harmonic oscillator (proof mass m, resonance frequency $\omega_0 = \sqrt{(\kappa/m)}$, quality factor Q) is given by the FDT applied to the mechanical susceptibility:

$$(14)\ S_x(\omega) = [4k_B T\omega_0 / (mQ)] \cdot [1 / ((\omega^2-\omega_0^2)^2 + \omega^2\omega_0^2/Q^2)]$$

Converting to equivalent acceleration noise spectral density (by multiplying by ω^4 and taking the low-frequency limit $\omega \ll \omega_0$), the acceleration noise floor is:

$$(15)\ a_n = \sqrt{(4k_B T\omega_0 / (mQ))} \quad [\mathrm{m/s^2/\sqrt{Hz}}]$$

The UBL variables are: $E_{scale} = k_B T$; $N_{freedom} \sim \omega_0/(2\pi Q)$ (the resonance linewidth, representing the thermally active bandwidth — a high-Q oscillator concentrates its thermal energy into fewer modes); $\kappa_{constraint} = mQ/\omega_0$ (the mechanical impedance of the damped oscillator, which is the physical parameter opposing fluctuations). Equation (15) shows that lower noise requires either lower temperature, larger mass, higher Q, or lower resonance frequency — a set of trade-offs fully captured by the UBL and of immediate relevance to inertial navigation, seismometry, and gravimetry [8].

3.6 Magnetic Johnson Noise

Just as conducting materials generate thermal voltage fluctuations (Johnson noise), they also produce fluctuating magnetic fields in their vicinity, because the thermally agitated conduction currents (Johnson noise currents) generate magnetic fields by Ampere's law. This magnetic Johnson noise is a fundamental limitation for sensitive magnetic sensors such as SQUIDs, fluxgates, and magnetoresistive magnetometers placed near conducting structures [9, 12]. For a conducting slab of thickness d, electrical conductivity σ_c, at a distance r from the sensor, the magnetic field noise spectral density scales as [12]:

$$(16)\ S_B(f) \approx \mu_o^2\, k_B T\, \sigma_c\, d\, /\, (8\pi\, r^2) \quad [\mathrm{T^2/Hz}]$$

The RMS magnetic field noise over bandwidth Δf is:

$$(17)\ B_{rms} \approx \sqrt{(\ \mu_o^2\, k_B T\, \sigma_c\, d\, \Delta f\, /\, (8\pi\, r^2)\ )}$$

The UBL structure is again evident: $E_{scale} = k_B T$; $N_{freedom} = \sigma_c\ d\ \Delta f$ (the product of conductance per unit area of the slab and the measurement bandwidth, representing the effective number of independent current-carrying modes that contribute magnetic field at the sensor location); $\kappa_{constraint} = 8\pi r^2\, /\, \mu_o^2$ (a geometric-permeability factor that quantifies how efficiently current fluctuations in the conductor couple magnetic field to the sensor). This is the precise magnetic analogue of Johnson noise in resistors, with the conductor's geometry and the vacuum permeability μ_o replacing electrical resistance.

3.7 Casimir Force Fluctuations

The Casimir effect [5] is among the most striking predictions of quantum field theory: two electrically neutral, perfectly conducting parallel plates in vacuum experience an attractive force per unit area arising from the modification of the zero-point energy spectrum of the electromagnetic vacuum by the conducting boundary conditions. The Casimir force per unit area at zero temperature is:

(18) $F/A = -\pi^2 \hbar c / (240\, d^4)$

where d is the plate separation, $\hbar$ is the reduced Planck constant, and c the speed of light [5, 14]. This force arises because the electromagnetic modes between the plates are quantized differently from those outside, and the zero-point energy $\hbar\omega/2$ of each modified mode contributes to the net force.

At finite temperature T, there are fluctuations in the Casimir force itself — variance in the force between the plates due to the statistical variation in thermal photon occupation numbers. A dimensional analysis of the thermal Casimir force fluctuation gives:

(19) $\delta F/A \approx \sqrt{(k_B T \cdot \hbar c / d^5)}$

Here the UBL variable identification reflects the hybrid quantum-classical nature of the Casimir regime. At low temperature ($k_B T \ll \hbar c/d$), the energy scale is $\hbar c/d$ (the characteristic zero-point photon energy for modes of wavelength ~ d); at high temperature, it is $k_B T$. The "freedom" parameter is the mode density ~ $1/d$ per unit area (the number of electromagnetic modes between the plates whose zero-point energy is significantly modified by the boundary), and the constraint is d^4 (the geometric factor governing the Casimir force itself). The UBL energy scale $E_{scale} = k_B T$ smoothly interpolates, via the full quantum Planck distribution, between these limiting regimes, and the resulting fluctuation amplitude is always of the square-root form of equation (3). For NEMS devices operating at sub-100 nm separations, these fluctuations can exceed thermomechanical noise — a crossover discussed further in Section 8 [14].

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4. Derivation of the Universal Baseline Law

4.1 Statement of the Law

We now state the Universal Baseline Law formally. For any physical observable q — displacement, voltage, magnetic field, force, or any generalized coordinate — in a system at or near thermal equilibrium (or quantum ground state), the irreducible baseline fluctuation (the minimum achievable RMS noise in q) is:

(20) $\sigma_q = \sqrt{(E_{\text{scale}} \cdot N_{\text{freedom}} / \kappa_{\text{constraint}})}$

where the three UBL variables are defined as follows:

- $\boldsymbol{E_{\text{scale}}}$ is the relevant energy scale per active mode. In the classical regime ($k_B T \gg \hbar\omega$): $E_{\text{scale}} = k_B T$. At the quantum zero-point limit ($T \to 0$): $E_{\text{scale}} = \hbar\omega/2$. In the general case, the full quantum FDT energy factor is $E_{\text{scale}} = (k_B T/2) \coth(\hbar\omega/2k_B T)$, which interpolates continuously between these limits.
- $\boldsymbol{N_{\text{freedom}}}$ is the number of thermally active, statistically independent degrees of freedom contributing to the observable q. For bandwidth-limited systems, $N_{\text{freedom}} \sim \Delta f/\Delta f_{\text{mode}}$; for single-mode systems, $N_{\text{freedom}} = 1$.
- $\boldsymbol{\kappa_{\text{constraint}}}$ is the generalized stiffness (restoring force per unit displacement in the generalized sense) conjugate to the observable q. For mechanical systems, it is the spring constant; for electrical systems, the inverse capacitance; for magnetic systems, the geometric-permeability factor; for RF systems, the characteristic impedance (per unit bandwidth).

4.2 General Derivation from the Partition Function

Consider a system with N independent degrees of freedom described by a purely quadratic Hamiltonian:

(21) $H(q_1, \ldots, q_N) = \sum_{i=1}^{N} (\kappa_i/2)\, q_i^2$

The canonical partition function at temperature T factorizes over independent modes:

(22) $Z = \prod_{i=1}^{N} \sqrt{(2\pi k_B T / \kappa_i)}$

From the Boltzmann distribution, the probability density for each coordinate is Gaussian: $P(q_i) = \sqrt{(\kappa_i/(2\pi k_B T))} \exp[-\kappa_i q_i^2/(2k_B T)]$. The variance of each mode is:

(23) $\langle q_i^2 \rangle = k_B T / \kappa_i$

Now consider a collective observable $Q = \sum_i q_i$. Since the modes are statistically independent (by the quadratic, non-interacting Hamiltonian), their variances add:

$$(24)\ \langle Q^2 \rangle = \sum_{i=1}^{N} k_B T / \kappa_i$$

If all modes share a common stiffness κ (or an effective stiffness obtained by harmonic mean over the mode distribution), equation (24) reduces to:

$$(25)\ \sigma_Q = \sqrt{(N k_B T / \kappa)}$$

This is equation (20) in its canonical form, with $N_{\text{freedom}} = N$ and $\kappa_{\text{constraint}} = \kappa$. The derivation is complete and rigorous for any system with a quadratic Hamiltonian. The law is thus a direct and necessary consequence of (a) Gaussian statistics for quadratic Hamiltonians, (b) the Boltzmann distribution, and (c) the statistical independence of normal modes. No additional assumptions are required.

4.3 Extension to Bandwidth-Limited Systems

Many physical systems of interest do not have a finite, discrete number of modes, but rather a continuous spectrum of modes with spectral density $S_q(f)$. In such systems, the observable's RMS fluctuation within a measurement bandwidth $[f_1, f_2]$ is:

$$(26)\ \sigma_q = \sqrt{(\int_{f_1}^{f_2} S_q(f)\, df)}$$

For a damped harmonic oscillator with quality factor $Q \gg 1$, the displacement spectral density is Lorentzian (equation 14) with half-power bandwidth $\Delta f = \omega_0/(2\pi Q)$ and on-resonance amplitude $S_x(f_0) = 4Qk_B T / (m\omega_0^3)$. Integrating over all frequencies:

$$(27)\ \sigma_x^2 = \int_0^\infty S_x(f)\, df = k_B T / \kappa$$

recovering equation (3) exactly, with no dependence on Q. This demonstrates that bandwidth integration over a Lorentzian spectral density always recovers the equipartition result — the total noise power is conserved regardless of how narrowly or broadly it is spectrally distributed, as required by the second law of thermodynamics. High-Q oscillators do not have less total noise; they concentrate it into a narrower bandwidth.

5. The Square-Root Structure and Its Geometric Origin

5.1 Variance vs. Amplitude: Why the Square Root Is Inescapable

It is a central point of the Universal Baseline Law that the physically measurable quantity — the RMS noise — scales as the *square root* of a ratio involving energy. This is not an accident of notation. The equipartition theorem constrains the mean energy stored in a degree of freedom: $\langle E \rangle = k_B T/2$. For a quadratic observable, the energy is $E = (\kappa/2)\sigma_q^2$. Setting these equal: $\sigma_q = \sqrt{(k_B T/\kappa)}$. The energy is a quadratic function of the amplitude; the amplitude is therefore necessarily the square root of the energy scaled by the stiffness. This is a kinematic identity — it follows from the definition of energy in terms of displacement, not from any dynamical assumption. Noise *power* is linear in temperature; noise *amplitude* is sublinear — it grows as $T^{1/2}$. Any claim that noise amplitude could scale linearly or otherwise in T would contradict the quadratic relationship between observable and energy that underlies all of classical and quantum mechanics near equilibrium.

5.2 The Phase-Space Ellipse

The geometric content of the UBL is most transparently displayed in phase space. For a single harmonic oscillator at temperature T, the joint probability distribution of position q and momentum p is the Boltzmann-weighted Gaussian:

(28) $P(q, p) \propto \exp[-\kappa q^2/(2k_B T) - p^2/(2mk_B T)]$

The 1σ contour of this distribution is an ellipse in phase space with semi-axes $\sigma_q = \sqrt{(k_B T/\kappa)}$ and $\sigma_p = \sqrt{(mk_B T)}$. The area enclosed by this ellipse is:

(29) $A_{1\sigma} = \pi\ \sigma_q\ \sigma_p = \pi\ k_B T\ /\ \omega_0$

where $\omega_0 = \sqrt{(\kappa/m)}$ is the oscillator frequency. The Heisenberg uncertainty principle requires $\sigma_q\ \sigma_p \geq \hbar/2$. The thermal area $\pi k_B T/\omega_0$ exceeds the quantum minimum $\pi\hbar/2$ by the factor $2k_B T/(\hbar\omega_0) \gg 1$ in the classical regime. As $T \to 0$, the thermal distribution collapses toward the quantum ground state, and the phase-space area approaches its minimum $\hbar/2$

— the zero-point fluctuation, which is the quantum instance of the UBL. The semi-axis σ_q sets the irreducible noise in position measurement; it is a direct geometric property of the thermal state in phase space.

5.3 Universality: Quadratic Hamiltonians as a Fixed Point

The square-root form is universal precisely because the quadratic Hamiltonian is the generic form near any stable equilibrium point. By definition, a stable equilibrium is a minimum of the potential energy. Expanding the potential in a Taylor series about the minimum, the linear term vanishes (at the minimum, all forces are zero), and the leading non-trivial term is quadratic. The quadratic approximation — the harmonic approximation — is therefore exact to leading order for any stable equilibrium in any physical system, in any domain. Only for extremely large fluctuations (exceeding the curvature radius of the potential) does anharmonicity become important, and only then does the square-root form break down. For all systems operating near thermal equilibrium at temperatures well below the barrier height, the UBL's square-root structure is an exact consequence of the harmonic approximation, not an empirical regularity.

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6. The Isotropy Argument

6.1 Isotropy of the Thermal Distribution in Phase Space

The Boltzmann distribution for a quadratic Hamiltonian (equation 28) is a multivariate Gaussian in the phase-space coordinates (q, p). When expressed in terms of the dimensionless, rescaled coordinates $\xi = q\sqrt{(\kappa/k_B T)}$ and $\pi = p / \sqrt{(m k_B T)}$, the distribution becomes:

(30) $P(\xi, \pi) \propto \exp[\,-(\xi^2 + \pi^2)/2\,]$

which is spherically symmetric (isotropic) in the rescaled phase space. The rescaling is precisely the one that normalizes out the physical parameters κ and m; in these natural units, no direction in phase space is distinguishable from any other. The thermal distribution neither "knows" nor "cares" about the distinction between position and momentum, or between voltage and charge — it simply populates all directions in the rescaled phase space equally. This is the phase-space expression of the second law of thermodynamics: equilibrium maximizes entropy, which for Gaussian distributions corresponds to maximum symmetry.

6.2 Consequence: Equipartition Is Exact and Inevitable

The isotropy of equation (30) in rescaled coordinates is mathematically equivalent to the statement of the equipartition theorem. Equal thermal energy in all rescaled directions means equal mean energy $\langle\xi^2\rangle/2 = \langle\pi^2\rangle/2 = 1/2$, which translates back to $\langle\kappa q^2/2\rangle = \langle p^2/(2m)\rangle = k_B T/2$. The UBL follows: in physical coordinates, the RMS of any observable is the inverse of the rescaling factor times the RMS in rescaled coordinates (which is unity): $\sigma_q = 1/\sqrt{(\kappa/k_B T)} = \sqrt{(k_B T/\kappa)}$. Different physical observables — voltage, displacement, field, force — correspond to different choices of observable coordinate, all of which project onto the same isotropic distribution in rescaled phase space. The UBL is therefore the statement that all such projections yield the same dimensionless RMS (unity in natural units), rescaled by the appropriate physical parameters.

6.3 Breaking Isotropy: Squeezing and the Heisenberg Trade-Off

The isotropy of the thermal distribution in phase space is a property of equilibrium. External fields, parametric driving, or quantum-optical squeezing operations can break this isotropy, redistributing the phase-space distribution from a circle to an ellipse in the natural units. In a squeezed state with squeezing parameter $r > 0$, the position noise is reduced below the UBL value:

(31) $\sigma_q \rightarrow \sqrt{(\hbar\, e^{-r} / (2\kappa))}$

while the conjugate momentum noise is amplified:

(32) $\sigma_p \rightarrow \sqrt{(\hbar\, \kappa\, e^{+r} / 2)}$

The product $\sigma_q \sigma_p = \hbar/2$ is preserved — saturating the Heisenberg uncertainty principle — but the isotropy in phase space is broken: the distribution is now an ellipse rather than a circle. The total phase-space area (and hence the total noise energy) is conserved. This is the only known mechanism to reduce noise below the UBL in one observable: squeezing transfers noise from a measured variable to its conjugate at the cost of increased uncertainty there. Squeezed states are employed in gravitational-wave detectors to push sensitivity below the standard quantum limit in the measurement quadrature [11, 15], at the cost of increased radiation-pressure noise in the other. The conservation of phase-space area thus demonstrates that the UBL expresses a conservation law — not merely a lower bound.

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7. Domain Mapping Table

Table 1 provides an explicit, quantitative mapping of each of the seven physical noise domains surveyed in Section 3 onto the three variables of the Universal Baseline Law (equation 20). The table is the central technical contribution of this paper. Each row specifies: the physical domain; the observable whose baseline fluctuation is characterized; the energy scale E_{scale}; the freedom parameter N_{freedom}; the constraint parameter $\kappa_{\text{constraint}}$; and the resulting baseline expression $\sigma_q = \sqrt{(E_{\text{scale}} \cdot N_{\text{freedom}} / \kappa_{\text{constraint}})}$.

The key assertion — which the table makes quantitatively precise — is that the UBL is not a metaphor, an analogy, or a dimensional-analysis heuristic. It is an exact identity: for each domain, the specific identification of the three UBL variables reproduces the known, experimentally verified physical result exactly under appropriate variable substitution. The table demonstrates that the structural similarity between noise formulas across domains is not superficial but reflects a deep, shared mathematical ancestry in the equipartition theorem and the fluctuation–dissipation theorem.

Domain	Observable σ	E_{scale}	$N_{freedom}$	$\kappa_{constraint}$	Baseline Expression
Johnson–Nyquist (resistor)	Voltage V_{rms}	$k_B T$	$4R\Delta f$	1	$\sqrt{(4k_B TR\Delta f)}$
Capacitor (kT/C)	Voltage V_{rms}	$k_B T$	1	$1/C$	$\sqrt{(k_B T / C)}$
RF Thermal Noise	Power P_N (or V_{rms})	$k_B T$	Δf	1 (power); $1/R$ (voltage)	$k_B T\Delta f$ (power); $\sqrt{(k_B TR\Delta f)}$ (voltage)
Brownian Motion	Displacement x_{rms}	$k_B T$	1 (harmonic); τ/τ_c (free)	κ_{spring}; $m\gamma$	$\sqrt{(k_B T/\kappa)}$; $\sqrt{(2k_B T\tau/m\gamma)}$
MEMS Accelerometer	Acceleration a_n	$k_B T$	$\omega_0/(2\pi Q)$	mQ/ω_0	$\sqrt{(4k_B T\omega_0/(mQ))}$ [per $\sqrt{\text{Hz}}$]
Magnetic Johnson Noise	B-field B_{rms}	$k_B T$	$\sigma_c d\Delta f$	$8\pi r^2/\mu_0^2$	$\sqrt{(\mu_0^2 k_B T\sigma_c d\Delta f/(8\pi r^2))}$
Casimir Fluctuation	Force/area $\delta F/A$	$\hbar c/d$ ($T\rightarrow 0$); $k_B T$ (high T)	Mode density ~ $1/d$ per unit area	d^4 (geometric)	$\sqrt{(k_B T \cdot \hbar c/d^5)}$ [thermal regime]

Notation: k_B *= Boltzmann constant;* T *= temperature;* R *= resistance;* Δf *= measurement bandwidth;* C *= capacitance;* κ *= spring constant;* m *= mass;* γ *= damping rate;* τ_c *= momentum relaxation time;* ω_0 *= resonance frequency;* Q *= quality factor;* σ_c *= electrical conductivity;* d *= conductor thickness or plate separation;* r *= sensor-conductor distance;* μ_0 *= vacuum permeability;* $\hbar$ *= reduced Planck constant;* c *= speed of light.*

The table makes a number of structural points explicit. First, the energy scale E_{scale} is $k_B T$ in six of the seven domains, confirming the classical thermal origin of noise floors in most practical systems. The Casimir domain is the sole case where the zero-point quantum energy $\hbar c/d$ serves as the primary energy scale at low temperature, but even there the high-temperature limit reverts to $k_B T$ via the Planck distribution. Second, the constraint $\kappa_{constraint}$ always has units of energy per (unit observable)^2 — it is the restoring stiffness in the generalized sense, regardless of the physical origin of the restoring mechanism. Third, the freedom parameter $N_{freedom}$ always enters linearly under the square root, so that increasing the number of active modes increases noise as $\sqrt{N_{freedom}}$ — a central design insight.

8. Implications and Applications

8.1 Fundamental Sensor Limits

The most immediate practical consequence of the UBL is the provision of a closed-form, irreducible lower bound on any sensor's noise floor. Equation (20) defines, for a given system and measurement condition, the minimum detectable signal that can in principle be achieved in thermal equilibrium. This bound applies with equal force to MEMS gyroscopes and accelerometers [8, 10], RF receivers [2], fluxgate and SQUID magnetometers [12], capacitive biosensors [9], and optomechanical force sensors [11, 15]. The minimum detectable signal is:

(33) $q_{\min} = \mathrm{SNR}_{\min} \times \sigma_q = \mathrm{SNR}_{\min} \times \sqrt{(k_B T\, N_{\mathrm{freedom}} / \kappa_{\mathrm{constraint}})}$

where $\mathrm{SNR}_{\min}$ is the minimum signal-to-noise ratio required for detection (typically 1 for RMS detection, or greater for reliable discrimination). The UBL bound is hard: no amount of electronic gain, digital filtering, or circuit optimization can reduce σ_q at fixed temperature and fixed system parameters. Reduction requires changes to the physical system: lower temperature ($T \downarrow$), narrower bandwidth ($N_{\mathrm{freedom}} \downarrow$), or greater stiffness ($\kappa_{\mathrm{constraint}} \uparrow$). This distinction between thermodynamic inevitability and engineering imperfection is operationally critical for sensor development programs.

8.2 Noise Engineering via the UBL

The UBL makes design trade-offs explicit and quantitative. Consider a capacitive MEMS pressure sensor as a representative example. The signal voltage generated by a pressure P applied to a capacitive element of gap capacitance C_0 and plate area A is $\Delta V \sim PA/C_0$. The noise voltage from equation (9) is $\sigma_V = \sqrt{(k_B T/C_0)}$. The signal-to-noise ratio is therefore:

(34) $\mathrm{SNR} = PA / \sqrt{(k_B T\, C_0)}$

Increasing C_0 (by reducing the gap or increasing the area) raises the signal but also raises the noise; the SNR scales as $C_0^{+1/2}$ via the A factor but also as $C_0^{-1/2}$ via the noise — a net scaling that depends on how A and C_0 are coupled by the geometry. Similarly, for a resistive sensor, the SNR trade-off between R (which amplifies signal but also noise via equation 8) is captured by the UBL. For a MEMS accelerometer (equation 15), reducing ω_0 (softer spring) lowers noise but also reduces the dynamic range and increases sensitivity to low-frequency environmental disturbances. Every such trade-off follows directly from the UBL and can be optimized by extremizing equation (34) or its analogue under the relevant constraints [8, 9].

8.3 The Quantum Limit and the Classical-to-Quantum Transition

As temperature is reduced toward zero, the energy scale in the UBL transitions from the classical $k_B T$ to the quantum zero-point energy $\hbar\omega_0/2$. This transition occurs at the crossover temperature $T^* = \hbar\omega_0/(2k_B)$. For a MEMS oscillator at $\omega_0/(2\pi) = 1$ MHz, $T^* \approx 24$ μK; for a microwave resonator at 10 GHz, $T^* \approx 240$ mK — readily accessible with dilution refrigerators. At $T \ll T^*$, the UBL noise floor becomes:

(35) $\sigma_q^{SQL} = \sqrt{(\hbar\omega_0 N_{\text{freedom}} / (2\kappa_{\text{constraint}}))}$

At this level, the Standard Quantum Limit (SQL) for continuous position measurement applies: the Heisenberg uncertainty principle prevents simultaneous precise measurement of q and its conjugate momentum, so measurement back-action introduces noise that is also bounded by the UBL. The combination of imprecision and back-action noise gives a total noise floor at the SQL: $\sigma_{\text{total}} \geq \sqrt{(\hbar/m\omega_0)}$, the standard quantum limit for force measurement [11]. LIGO gravitational-wave detectors operate in this regime and employ squeezing (Section 6.3) to circumvent the SQL in one quadrature [15].

8.4 Cross-Domain Noise Budgeting

In practical systems where multiple noise mechanisms are simultaneously active — a MEMS accelerometer that also has Johnson noise in its readout electronics, thermal magnetic noise from surrounding conductors, and potentially Casimir fluctuations in next-generation devices — the UBL provides a unified framework for noise budgeting. Since all noise contributions are uncorrelated (independent physical mechanisms), the total observable noise is:

(36) $\sigma_{total} = \sqrt{(\sigma_{mechanical}^2 + \sigma_{electrical}^2 + \sigma_{magnetic}^2 + \sigma_{Casimir}^2 + \ldots)}$

where each σ_i is given by the UBL with the appropriate domain parameters. The dominant noise contribution is the one with the largest ratio $E_{scale} \cdot N_{freedom} / \kappa_{constraint}$, and noise budgeting reduces to comparing these ratios across domains — a straightforward task made possible by the unified UBL framework. The cross-domain noise budget is particularly valuable in quantum sensor design, where the system must be engineered so that no single noise mechanism dominates — a condition sometimes called "balanced noise budget" in the MEMS literature [10].

8.5 Implications for the Casimir Regime in NEMS

At nanometer-scale separations in next-generation NEMS (nanoelectromechanical systems) devices, Casimir force fluctuations (equation 19) can exceed thermomechanical displacement noise. The crossover condition, derived by equating the Casimir fluctuation force to the thermomechanical noise force, is:

(37) $\sqrt{(k_B T \cdot \hbar c/d^5)} \sim \sqrt{(4k_B T\omega_0/(mQ))} \times \kappa$

Solving for the critical separation d_c below which Casimir fluctuations dominate:

(38) $d_c \sim (\hbar c \cdot mQ / (4\kappa^2 \omega_0))^{1/5}$

For typical silicon NEMS parameters ($m \sim 10^{-15}$ kg, $\kappa \sim 1$ N/m, $Q \sim 10^4$, $\omega_0/(2\pi) \sim 10$ MHz), this gives $d_c \sim 10$–50 nm [14]. NEMS switches and oscillators with gap sizes in this range will be Casimir-noise-limited rather than thermomechanically limited at room temperature — a design constraint made quantitatively explicit by the UBL for the first time.

9. Conclusion

We have demonstrated that the noise floors of seven superficially distinct physical systems — Johnson–Nyquist resistor noise, kT/C capacitor noise, RF thermal noise, Brownian motion, MEMS thermomechanical noise, magnetic Johnson noise, and Casimir force fluctuations — are all instances of a single, exact law: the Universal Baseline Law,

(39) $\sigma_q = \sqrt{(E_{\text{scale}} \cdot N_{\text{freedom}} / \kappa_{\text{constraint}})}$

The law derives rigorously and directly from the equipartition theorem and the fluctuation–dissipation theorem. Its square-root structure is forced by the quadratic relationship between physical observables and energy — a universal property of any stable equilibrium in classical or quantum mechanics — and is not an approximation. The geometric interpretation via the thermal phase-space distribution (Sections 5 and 6) demonstrates that the UBL is the expression of a single underlying symmetry: the isotropy of the Boltzmann distribution in rescaled phase-space coordinates. The domain mapping table (Table 1) provides explicit, quantitative correspondences that reproduce all seven known physical noise results exactly under appropriate variable substitution.

The practical significance of the UBL is substantial. It provides a universal, domain-agnostic lower bound on sensor noise; it guides noise engineering by making design trade-offs explicit; it unifies cross-domain noise budgeting; it predicts the classical-to-quantum crossover temperature for any physical system; and it identifies the critical separation at which Casimir force fluctuations dominate thermomechanical noise in NEMS devices. In each case, the UBL answers the practical question: "What is the irreducible noise, and how does it depend on my design parameters?" with a single, closed-form expression.

This work opens several directions for future investigation. The UBL mapping should be extended to optical phase noise (where the "constraint" is the cavity finesse and the "freedom" is the photon number fluctuation), quantum amplifier added noise (where the quantum FDT gives the half-photon zero-point noise floor of any phase-preserving amplifier), and non-equilibrium fluctuation theorems (Jarzynski, Crooks), where the energy scale E_{scale} is replaced by a free-energy difference. Most intriguingly, the UBL may admit a deeper information-theoretic interpretation: the connection between σ_q and the Cramér–Rao bound of classical estimation theory suggests that the UBL may express a fundamental limit on the extractable Fisher information per degree of freedom in a thermal system — a direction that would connect noise physics to quantum information theory at the most fundamental level.

References

1. [1] J. B. Johnson, "Thermal Agitation of Electricity in Conductors," *Physical Review*, vol. 32, pp. 97–109, July 1928.
2. [2] H. Nyquist, "Thermal Agitation of Electric Charge in Conductors," *Physical Review*, vol. 32, pp. 110–113, July 1928.
3. [3] A. Einstein, "Über die von der molekularkinetischen Theorie der Wärme geforderte Bewegung von in ruhenden Flüssigkeiten suspendierten Teilchen," *Annalen der Physik*, vol. 17, pp. 549–560, 1905.
4. [4] H. B. Callen and T. A. Welton, "Irreversibility and Generalized Noise," *Physical Review*, vol. 83, no. 1, pp. 34–40, July 1951.
5. [5] H. B. G. Casimir, "On the Attraction Between Two Perfectly Conducting Plates," *Proceedings of the Koninklijke Nederlandse Akademie van Wetenschappen*, vol. 51, pp. 793–795, 1948.
6. [6] R. Kubo, "The Fluctuation-Dissipation Theorem," *Reports on Progress in Physics*, vol. 29, pp. 255–284, 1966.
7. [7] L. Boltzmann, *Vorlesungen über Gastheorie*. Leipzig: J. A. Barth, 1896.
8. [8] T. B. Gabrielson, "Mechanical-Thermal Noise in Micromachined Acoustic and Vibration Sensors," *IEEE Transactions on Electron Devices*, vol. 40, no. 5, pp. 903–909, May 1993.
9. [9] S. Kogan, *Electronic Noise and Fluctuations in Solids*. Cambridge: Cambridge University Press, 1996.

10. [10] T. Sleator, E. L. Hahn, C. Hilbert, and J. Clarke, "Thermomechanical noise floor in sensor design," *Journal of Applied Physics*, vol. 68, pp. 508–512, 1990.

11. [11] V. B. Braginsky and F. Ya. Khalili, *Quantum Measurement*. Cambridge: Cambridge University Press, 1992.

12. [12] P. R. Saulson, "Thermal noise in mechanical experiments," *Physical Review D*, vol. 42, no. 8, pp. 2437–2445, Oct. 1990.

13. [13] S. A. Diddams, J. C. Bergquist, S. R. Jefferts, and C. W. Oates, "Standards of Time and Frequency at the Outset of the 21st Century," *Science*, vol. 306, no. 5700, pp. 1318–1324, Nov. 2004.

14. [14] M. Bordag, G. L. Klimchitskaya, U. Mohideen, and V. M. Mostepanenko, *Advances in the Casimir Effect*. Oxford: Oxford University Press, 2009.

15. [15] C. M. Caves, "Quantum-Mechanical Noise in an Interferometer," *Physical Review D*, vol. 23, no. 8, pp. 1693–1708, Apr. 1981.

16. [16] L. D. Landau and E. M. Lifshitz, *Statistical Physics*, 3rd ed. Oxford: Pergamon Press, 1980.

17. [17] R. Brown, "A Brief Account of Microscopical Observations Made in the Months of June, July and August 1827, on the Particles Contained in the Pollen of Plants; and on the General Existence of Active Molecules in Organic and Inorganic Bodies," *Philosophical Magazine*, vol. 4, pp. 161–173, 1828.